MedAssist AI

Medical Symptom Analysis & Disease Prediction System

MILESTONE 3 — FRONTEND DEVELOPMENT & PATIENT MODULE INTEGRATION

Progress documented up to 09 August 2026

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Internship	Infosys Springboard Internship	**Project Phase**	Frontend + End-to-End Integration

1. Milestone 3 Objective

The objective of Milestone 3 was to convert the completed backend foundation into a usable React-based healthcare application. The focus was on building the patient-facing interface, connecting frontend screens with FastAPI APIs, implementing the complete patient workflow, and providing a professional, responsive user experience.

Milestone 2 established the backend foundation including PostgreSQL, authentication, disease prediction, risk assessment, recommendations, health reports, medical history, profile management and analytics APIs. ■filecite■turn15file0■

2. Frontend Development Progress

Frontend Area	Progress	Implementation
Landing Page	**Completed**	Hero section, About, CTA, navigation and accessible patient actions.
Registration & Login	**Completed**	React forms, navigation and backend authentication integration.
Patient Dashboard	**Completed**	Sidebar, navbar, patient information, quick actions and responsive layout.
Symptom Analysis	**Completed**	Three-step symptom workflow with organized symptom selection and navigation.
Disease Prediction	**Completed**	Prediction result, confidence information and continuation to risk assessment.
Risk Assessment	**Completed**	Risk score, risk level, health factors and treatment continuation workflow.
Treatment Recommendation	**Completed**	Disease description, medicines, diet, workout, precautions and doctor advice.

Milestone 3 — Patient Module Progress

The patient module has been developed as an integrated workflow rather than as isolated screens. The current frontend connects the major healthcare journey from symptom entry through prediction, risk assessment, recommendations, reports, history and analytics.

Patient Module	Status	Current Progress
Health Report	Completed	Professional report view and report generation/download flow integrated.
Medical History	Completed	Historical predictions/reports are displayed for patient review.
Patient Analytics	Completed	Disease/risk insights and prediction trends are presented through the dashboard.
Settings / Profile	Completed	Profile details, password management and profile photo upload/update UI implemented.
Patient Navigation	Completed	Sidebar and protected patient routes connect the major patient pages.
Responsive UI	Completed	Bootstrap-based responsive layout and reusable dashboard styling implemented.
End-to-End Validation	In Progress	Final workflow validation and edge-case fixes are being performed.

3. Key Frontend Features Implemented

- **React + Bootstrap UI:** The application interface was developed using React components and Bootstrap-oriented responsive layouts.
- **Patient Dashboard:** A structured dashboard provides access to symptom analysis, disease prediction, medical history, reports, analytics and settings.
- **Three-Step Symptom Analysis:** Symptoms are selected through a guided multi-step interface instead of a single crowded form.
- **Prediction-to-Risk Workflow:** The predicted disease is carried into risk assessment and then into the treatment recommendation stage.
- **Recommendation Experience:** Recommendation results are separated into meaningful sections including description, medicines, diet, workout, precautions and medical advice.
- **Health Reports:** Patient report screens are connected to the backend report-generation functionality.
- **Profile Management:** Patient profile fields, security settings and profile-photo upload functionality are available through Settings.
- **Navigation and Access:** Landing-page CTA buttons now provide direct access to patient registration/login actions, while dashboard navigation supports the authenticated workflow.

4. Backend–Frontend Integration

The React frontend communicates with the FastAPI backend through HTTP requests. The integrated workflow uses backend services for authentication, patient profile retrieval/update, disease prediction, risk assessment, recommendations, reports, medical history and analytics. The backend architecture and API modules were already established in Milestone 2. ■filecite■turn15file0■

Layer	Technology / Component
Frontend	React.js, React Router, Bootstrap, React Icons, Axios

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Backend	FastAPI, Python, REST APIs
Database	PostgreSQL with SQLAlchemy ORM
Machine Learning	Scikit-learn, Random Forest, joblib, label encoder
Security	JWT authentication and BCrypt password hashing
Reports	ReportLab-based PDF report generation
Testing / Development	Swagger UI, VS Code, Git and GitHub

Milestone 3 — Status, Validation & Next Phase

Milestone 3 has significantly advanced MedAssist AI from a backend-centered implementation to a complete patient-facing web application. The main patient journey is now represented in the frontend, with the remaining work focused on final validation, bug fixing, UI polish and preparation for the administrator module.

5. Current Overall Project Status

Area	Status	Notes
Datasets & Preprocessing	Completed	Medical datasets collected, cleaned and prepared for prediction/recommendation workflows.
Machine Learning	Completed	Disease prediction model and label encoder integrated with backend prediction service.
FastAPI Backend	Completed	Core patient APIs and healthcare workflow services implemented.
PostgreSQL Database	Completed	Relational database and ORM models established for patient and healthcare records.
Patient Frontend	Completed	Major patient screens and workflow integration implemented.
Patient End-to-End Testing	In Progress	Final validation and issue resolution underway.
Admin Frontend Module	Next Phase	To be developed after patient-module validation is finalized.

6. Known Issue Under Validation

Duplicate prediction/report records: During medical-history validation, a single patient prediction action has been observed to result in duplicate records/reports in some cases. The prediction service itself can return one prediction, but the persistence/report workflow is being checked to identify why the same operation is creating two database entries or two reports.

This issue is being treated as a final patient-module validation item rather than as a new feature. The next debugging step is to trace the frontend API calls, backend prediction/report creation flow, and database insert operations to ensure one user action creates one prediction ID and one corresponding report.

7. Next Steps

- **Finalize patient module:** Resolve duplicate prediction/report creation and complete end-to-end validation.
- **Verify database consistency:** Confirm prediction, report and medical-history relationships and prevent unintended duplicate inserts.
- **Complete admin module:** Build the administrator dashboard and connect patient/report/risk monitoring features.
- **Admin analytics:** Integrate disease distribution, risk distribution, prediction trends, common diseases and common symptoms.
- **System testing:** Perform complete frontend-backend testing across registration, login, symptom analysis, prediction, risk, recommendations, reports, history, analytics and settings.
- **Final documentation:** Prepare final project documentation, screenshots, testing evidence and presentation material.
- **Deployment preparation:** After validation, prepare the application for deployment and final demonstration.

8. Conclusion

Milestone 3 successfully advances MedAssist AI into a functional patient-facing healthcare application. The frontend now covers the major patient journey and is integrated with the previously completed backend services. The work has moved from core implementation toward validation, quality improvement and final system integration. The immediate priority is to complete patient-module validation, after which development will proceed to the administrator module.

Project direction: Patient Module → Final Validation → Admin Module → Complete System Testing → Deployment & Final Demonstration